The Zeta Function of the T^4/Z_3 Torus:

Orbifold Projection, Dirichlet L-Functions and Hecke Structure of the GF(27) Mass Layers

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Abstract

The Galois factorisation classes of Doc. 342 and the harmonic series on the T^4 torus of Doc. 159 together yield a precise statement about the zeta function of the FFGFT spacetime.

Four results: **(A)** The Z_3 orbifold projection of T^4/Z_3 transforms the full Riemann zeta into a Dirichlet L-function: $\zeta_{T^4/\mathbb{Z}_3}(s) = L(s, \chi_0) = (1 - 3^{-s})\,\zeta(s)$ **[B]**. **(B)** The Euler product of this L-function is hierarchically ordered by the entry depth $k = \mathrm{ord}_p(3)$; the physically visible primes $\{2, 3, 5, 7, 11, 13\}$ are exactly those with $k \leq 6$ **[B]**. **(C)** The orbifold projection generates additional zeros at $s = 2\pi ik/\ln 3$ ($k \in \mathbb{Z} \setminus \{0\}$) on the imaginary axis $\mathrm{Re}(s) = 0$; these are the algebraic signature of the Z_3 sector separation **[B]**. **(D)** The four Frobenius conjugacy classes of primitive elements in $\mathrm{GF}(27)^*$ (Doc. 342, Theorem A) correspond to four Hecke L-functions for the characters of $\mathrm{Gal}(\mathrm{GF}(27)/\mathrm{GF}(3)) \cong \mathbb{Z}_3$; whether these four classes carry distinct mass layers remains open **[S]**.

Check script: `pruef_343_zeta_galois.py`.

.1 Status markers and starting point

Status markers

- **[K]** — *Confirmed*: fully proved from ξ or a corpus-established derivation; all check-script assertions passed.
- **[B]** — *Established*: numerically secured; analytical proof present or trivial.
- **[S]** — *Speculative*: indication or conjecture, not yet fully proved.

Doc. 159 shows that the spectral zeta function of the Laplace operator on a torus T^d is the Epstein zeta function, whose one-dimensional special case is the Riemann zeta $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$. Doc. 342 shows that primes appear in the GF(3^k) hierarchy of the T^4/Z$_3$ orbifold ordered by entry depth $k = \mathrm{ord}_p(3)$. This document combines both: what does the GF(3^k) hierarchy imply for the zeta function itself?

.2 Theorem A: Orbifold projection and Dirichlet L-function

Decomposition of the zeta by Z$_3$ classes

On T^4/Z$_3$ the allowed modes $n \in \mathbb{Z}^4$ are structured by Z$_3$ invariance. In the one-dimensional analogue $\zeta(s)$ splits into three partial sums by n mod 3 **[B]**:

$$\zeta(s) = \underbrace{\sum_{\substack{n=1\\3|n}}^{\infty} \frac{1}{n^s}}_{=3^{-s}\zeta(s)} + \underbrace{\sum_{\substack{n=1\\n\equiv1(3)}}^{\infty} \frac{1}{n^s}}_{L_1(s)} + \underbrace{\sum_{\substack{n=1\\n\equiv2(3)}}^{\infty} \frac{1}{n^s}}_{L_2(s)} \tag{1}$$

The fixed-point sector ($n \equiv 0$) contributes exactly $3^{-s}\zeta(s)$; the twist sectors ($n \equiv \pm1$) together $(1-3^{-s})\zeta(s)$.

Principal character χ_0 mod 3

The Dirichlet character χ_0 mod 3 with $\chi_0(n) = 0$ if $3 \mid n$, otherwise $\chi_0(n) = 1$, gives **[B]**:

$$L(s,\chi_0) = \sum_{\mathrm{gcd}(n,3)=1} \frac{1}{n^s} = (1-3^{-s})\,\zeta(s) = \prod_{p\neq3} \frac{1}{1-p^{-s}} \tag{2}$$

This is the **spectral zeta function of the T^4/Z$_3$ torus**: the characteristic prime $p = 3$ does not appear as an Euler factor — it is the geometric foundation of the orbifold and stands outside the prime hierarchy.

Non-trivial character χ_1 mod 3

The character χ_1 mod 3 with $\chi_1(1) = +1$, $\chi_1(2) = -1$, $\chi_1(0) = 0$ measures the *asymmetry* of the twist sectors **[B]**:

$$L(s,\chi_1) = \sum_{\substack{n=1\\n\equiv1(3)}}^{\infty} \frac{1}{n^s} - \sum_{\substack{n=1\\n\equiv2(3)}}^{\infty} \frac{1}{n^s} \tag{3}$$

with the exact value (Dirichlet 1837): $L(1,\chi_1) = \pi/(3\sqrt{3})$.

Physical meaning **[S]**: $L(s,\chi_1)$ measures the difference between winding numbers $w = +1$ and $w = -1$ on the torus, i.e. the Z$_3$ sector asymmetry. Whether $L(s,\chi_1)$ has a direct physical observable in FFGFT (particle–antiparticle asymmetry in the twisted sector?) remains open **[S]**.

.3 Theorem B: Hierarchical Euler product by $k = \mathrm{ord}_p(3)$

The primes in $L(s,\chi_0) = \prod_{p\neq 3}(1-p^{-s})^{-1}$ are not equivalent. By the entry-depth rule $k = \mathrm{ord}_p(3)$ from Doc. 342, Theorem B, the Euler product is hierarchically ordered **[B]**:

$$L(s,\chi_0) = \prod_{k=1}^{\infty} \prod_{\substack{p\neq 3 \\ \mathrm{ord}_p(3)=k}} \frac{1}{1-p^{-s}} \tag{4}$$

Table 1: Hierarchy of Euler factors by $k = \mathrm{ord}_p(3)$

k	new prime(s)	Euler factor at $s = 2$	harmonics	FFGFT role
1	2	$4/3 = 1.333$	octave	Z_2 fixed point, massive sector
3	13	≈ 1.006	thirteenth	GF$(27)^*$, leptons
4	5	≈ 1.042	major third	factor in $(m_\mu/m_e)^2$
5	11	≈ 1.008	eleventh	neutrinos (Doc. 340)
6	7	≈ 1.021	seventh	top quark (Doc. 289)
≥ 7	$\geq 41, 1093, \dots$	$\approx 1 + p^{-2}$	no small harmonics	suppressed

Fractal suppression of higher Euler factors

On the fractal T^4 torus (Doc. 159) high-mode contributions are suppressed by the damping factor $\tau^{-(1+\xi)k}$. This transfers to the Euler factors; the *weighted* Euler product reads

$$\zeta^{\text{frac}}_{T^4/\mathbb{Z}_3}(s) \;\cong\; \prod_{k=1}^{\infty} \prod_{\substack{p\neq 3 \\ \mathrm{ord}_p(3)=k}} \left(\frac{1}{1-p^{-s}}\right)^{\tau^{-k(1+\xi)}} \tag{5}$$

Since $\tau^{-k(1+\xi)} \to 0$ as $k \to \infty$, each Euler factor is pushed toward 1 for large k: primes with $\mathrm{ord}_p(3) \geq 7$ make no physical contribution **[B]**. This is Theorem D of Doc. 342 in the language of the zeta function.

.4 Theorem C: New zeros from the orbifold projection

Zeros of $(1-3^{-s})$

The factor $(1-3^{-s})$ in $L(s,\chi_0) = (1-3^{-s})\zeta(s)$ has zeros at **[B]**:

$$3^{-s} = 1 \quad\Longleftrightarrow\quad s = \frac{2\pi i\, k}{\ln 3}, \qquad k \in \mathbb{Z}\setminus\{0\} \tag{6}$$

These lie on $\mathrm{Re}(s) = 0$ (imaginary axis), *not* on the critical line $\mathrm{Re}(s) = 1/2$ of the Riemann hypothesis. They are absent from the Riemann zeta spectrum; they arise solely from the Z_3 orbifold projection.

The zero spacing $2\pi/\ln 3 \approx 5.72$ encodes the periodicity of the Z_3 lattice structure. A resonance test of the first eight known Riemann zeros γ_n against this unit shows no significant coincidence (deviations 0.15–0.47) **[B]**: the full Riemann zeta encodes the distribution of *all* primes, not only the Z_3-compatible ones.

Complete zero structure of the orbifold zeta

$L(s, \chi_0)$ has three types of zeros:

- **Trivial zeros of $\zeta(s)$**: at $s = -2, -4, -6, \ldots$
- **Non-trivial Riemann zeros**: on $\mathrm{Re}(s) = 1/2$ (assumed), unchanged
- **Orbifold zeros**: at $s = 2\pi i k / \ln 3$, $k \neq 0$ — new, generated by the Z_3 projection **[B]**

.5 Theorem D: Four Hecke L-functions of the GF(27) structure

Galois group and its characters

The Galois group Gal(GF(27)/GF(3)) $\cong \mathbb{Z}_3$ has three characters: the trivial χ_{triv} and two non-trivial χ_ω, χ_{ω^2} with $\omega = e^{2\pi i/3}$ **[B]**.

The four Frobenius conjugacy classes of primitive elements in GF$(27)^*$ (Doc. 342, Theorem A: polynomials f_1, f_2, f_3, f_4) correspond to four Hecke L-functions describing the splitting behaviour of primes in the extension GF(27)/GF(3):

Table 2: Splitting of primes $p \neq 3$ in GF(27)/GF(3)

Splitting type	Condition	Hecke L-function
totally split	$p \equiv 1 \pmod 3$, $\mathrm{ord}_p(3) \mid 1$	trivial χ_{triv}
3-cycle	$\mathrm{ord}_p(3) = 3$ (e.g. $p = 13$)	$L(s, \chi_\omega)$
3-cycle	$\mathrm{ord}_p(3) = 3$	$L(s, \chi_{\omega^2})$
inert	$\mathrm{ord}_p(3) \nmid 3$	no contribution

Open question

Theorem A of Doc. 342 assigns the four primitive cubics $f_1, \ldots, f_4$ to the four Frobenius conjugacy classes in GF$(27)^*$. Whether these four classes carry physically distinct mass layers (analogous to f_1 carrying the lepton layer) is an open question **[S]**. If so, each layer would be characterised by its own Hecke L-function.

.6 Summary

Table 3: Results and status

Theorem	Content	Status
A	$\zeta_{T^4/\mathbb{Z}_3}(s) = (1 - 3^{-s})\zeta(s) = L(s, \chi_0)$	**[B]**
A′	$L(s, \chi_1)$ measures twist-sector asymmetry	**[S]**
B	Euler product hierarchical by $k = \mathrm{ord}_p(3)$	**[B]**
B′	Fractal damping suppresses Euler factors $k \geq 7$	**[B]**
C	Orbifold zeros at $s = 2\pi i k / \ln 3$ on $\mathrm{Re}(s) = 0$	**[B]**
C′	No resonance of Riemann zeros with Z_3 unit	**[B]**
D	Four Hecke L-functions for Gal(GF(27)/GF(3))$\cong \mathbb{Z}_3$	**[B]**
D′	Physics of the four layers $f_1, \ldots, f_4$	**[S]**

The central statement: the full Riemann zeta $\zeta(s)$ belongs to the flat $\mathbb{R}^4$ spectrum. The zeta of the T^4/Z_3 orbifold is $L(s,\chi_0) = (1-3^{-s})\,\zeta(s)$ — a filtered version in which the 3-Euler factor is separated out and the primes are weighted by the Galois hierarchy $k = \mathrm{ord}_p(3)$. This is the zeta-function form of the entry rule of Doc. 342.

Check scripts

Directory `2/python/Dok343_Skripte/`, wrapper `run_all_343.sh`:

- `pruef_343_zeta_galois.py` — Theorems A–D; partial zeta decomposition, L-functions, zeros, resonance test, weighted Euler product, Hecke discussion.

All assertions passed.

Note on the use of AI assistance

This document was prepared with the assistance of Claude (Anthropic). All numerical computations are verified by Python check scripts. The physical interpretations and conclusions are those of Johann Pascher (ORCID 0009-0000-6518-4064).

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